CARBON MANAGEMENT TECHNOLOGY CONFERENCE

CMTC 151756

CO_2 Sequestration in Deepwater Sediments Offshore Japan

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This paper was prepared for presentation at the Carbon Management Technology Conference held in Orlando, Florida, USA, 7–9 February 2012.

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Abstract

When injected in deep saline aquifers or depleted oil and gas reservoirs, supercritical CO_2 remains mobile and can, therefore, migrate through any conduits or fractures. In addition, public opinion, regulations and the lack of space for CO_2 injection in some densely populated regions of the world such as the Japanese archipelago encourage investigating other alternatives such as carbon dioxide sequestration in deepwater sub-seabed formations.

This paper intends to present a technical feasibility study of CO_2 sequestration in deepwater sediments offshore Japan. The main processes, technical requirements, technologies and structures that are currently available to transport and inject liquid CO_2 successfully in sub-seabed formations below 9,000 feet of water (≈2,750 meters) are first discussed. Then, three storage sites situated offshore Japan; one located in the Northwest Pacific Ocean near the island of Shikoku; another located in the Sea of Japan near the island of Honshu; and the last one located farther in the Northwest Pacific Ocean in ultra-deepwater (18,000 feet); are selected to conduct reservoir simulations.

From this study, it appears that CO_2 capturing technologies and transporting means seem to be at a mature stage. Also, current and newest 5th and 6th generation drilling vessels are estimated to be capable of drilling very shallow wells in water depths greater than 9,000 feet and even in ocean waters as deep as 18,000 feet if new materials such as titanium or composite for riser systems were to be deployed for both the drilling and CO_2 injection operations. However, CO_2 storing and injection facilities are not available yet to unload large quantities of CO_2 offshore. As a result, some concepts should be designed, qualified and tested for these large scale operations within the next decade to demonstrate through pilot projects the technical feasibility of CO_2 sequestration in sub-seabed geological formations.

Additionally, the main findings from this comparative study and reservoir simulations conducted at three different sites located offshore Japan confirm that a significant part of ultra-deepwater regions with at least 9,000 feet of ocean water and planar seafloor are appropriate for CO_2 storage. Secondly, reservoir models confirm that due to high pressures and low temperatures reigning at water depths greater than 9,000 feet, the liquid CO_2 injected in the first few hundred feet of sediments has a higher density than the surrounding formation pore-fluid and therefore remains buoyantly trapped under certain condition of geothermal gradient, sediments permeability and formation pressure and; hence constitute a valid and safe CO_2 storage candidate.

Introduction

CO_2 annual emissions currently account for about 30 billion tons (IEA, IPCC, 2010). In addition, carbon dioxide released in the atmosphere in the near future, by 2030, is expected to reach at least 40 billion tons emitted annually. Even though almost half of this greenhouse gas will eventually be absorbed by natural processes (CO_2 slowly moved from surface waters of the

Earth's oceans and seas to deeper waters because the solubility of CO_2 in water increases with increasing pressure), the remaining part of the CO_2 may stay in the Earth's atmosphere for centuries (Marchetti, 1977).

Besides, if one compares the current annual emission of CO_2 with the largest offshore carbon sequestration project in the world where about 1 million tons of CO_2 has been injected each year for the last 15 years in the permeable Utsira sandstone formation in Norway, it becomes evident that a very large number (greater than 10,000) of carbon sequestration projects utilizing multiple approaches and technologies would be needed to stabilize atmospheric CO_2 emission levels.

To date, the bulk of CO_2 sequestration studies and pilot projects have focused on capturing the CO_2 and injecting it, usually in a supercritical state, in deep saline aquifers, depleted oil and gas fields or abandoned coal mines where a physical trap would not allow, in principle, any leakage of the injected CO_2. However, at the relatively high pressures and high temperatures present at these formation depths, the supercritical CO_2 becomes buoyant and therefore, can migrate if the formation is not perfectly sealed. Therefore, any already present micro-fractures or rock property changes due to excessive CO_2 injection pressure or regional tectonics could represent a potential leakage mechanism for the buoyant CO_2.

In fact, governments and public opinion would probably be opposed to extensive drilling campaign for injection wells and CO_2 sequestration operations taking place at injection sites located at close distance from populated areas because of the potential risks of having the CO_2 leaking and migrating up the ground surface. Thus, in some densely populated regions of the world such as the Japanese archipelago which has very little available space for large onshore CO_2 sequestration projects, other alternatives such as injecting and sequestering carbon dioxide in offshore formations must be investigated. In addition, since there is an intense tectonic activity in Japan, the risk of having buoyant supercritical CO_2 leaking from the geologic formations where it has been injected into is too great and therefore would require a different type of trapping mechanism for the CO_2 such as a gravitational trap where the heavier fluid (i.e. liquid CO_2) would remain below the lighter fluid (i.e. sediments pore-fluid) even though geomechanical perturbations occur.

Since all carbonate species are very dependent on temperature and pressure conditions, other CO_2 sequestration methodologies had to be investigated in order to find a pressure and temperature window where carbon dioxide could be safely and permanently stored underground. Thus, based on theoretical and experimental works that have taken place over the last three decades to study whether the Earth's oceans and marginal seas could sequester anthropogenic CO_2 (Nakashiki et al, 1995), it has been found that when liquid CO_2 is injected at great water depths, beyond 9,000 ft (≈2,750 meters), CO_2 actually becomes denser than seawater because of the low temperatures (between 35°F and 40°F) and high pressures (greater than 4,000 psi) reigning at these water depths (Fer and Haugen, 2003). However, because deep ocean currents will eventually transport some of the CO_2, which has not dissolved with seawater, to the water surface and eventually release it back into the atmosphere, a different environment, current-free and with a better trapping mechanism is therefore required to permanently store the CO_2.

Thus, it has recently been proposed that CO_2 could be injected just below the seafloor and benefit from the unique pressure and temperature regimes occurring at these great water depths (House et al, 2006 and 2008, Schrag, 2009). Moreover, since the International Maritime Organization announced in 2007 that storage of CO_2 under the seabed was allowed, injecting and storing CO_2 in deepwater sediments appears as one of the safest and most viable CO_2 sequestration option.

In this paper, three ultra-deepwater storage sites with water depths ranging between 9,300 ft and 18,200 ft situated offshore Japan; one located in the Pacific Ocean near the island of Shikoku; another located in the Sea of Japan near the island of Honshu and; the last one located farther in the Northwest Pacific Ocean (N40° E150°), are selected to conduct reservoir simulations. Also, the main processes, technical requirements, technologies and structures that are currently available to transport and inject liquid CO_2 successfully in deepwater sub-seabed formations are reviewed.

Potential CO_2 Storage Sites Offshore Japan

The Nippon Foundation and the General Bathymetric Chart of the Oceans (GEBCO) have collaborated and published in 2004 a world bathymetric map. This effort resulted in a high resolution world seafloor map based upon data from GEBCO one minute grid (GEBCO, 2003) and continent topographic map taken from the U.S. National Aeronautics and Space Administration Blue Marble data set (NASA, 2002). This high resolution bathymetric map showed and confirmed that a large

part of the world's oceans and seas is deeper than 10,000 feet (depths greater than 3 kilometers are colored gradually from true blue to dark blue on Figure 1).

In addition, it is important to note that some deep water areas are located relatively close to the main clusters of CO_2 emissions with the metropolis of Yokohama-Tokyo, Saitama and Sendai located close to the Northwest Pacific Basin deepwater sites and the cities of Sapporo, Nagoya, Osaka, Kobe and Kyoto located closer to the Japan and Yamato Basin deepwater areas (as shown on Figure 1).

However, studies of sub-seabed properties are necessary to ensure that the sediments have the right physical (porosity, permeability, thermal conductivity) and chemical properties (reactivity, stability) and a minimum thickness to sustain large quantities of CO_2 being injected in those sub-seabed formations.

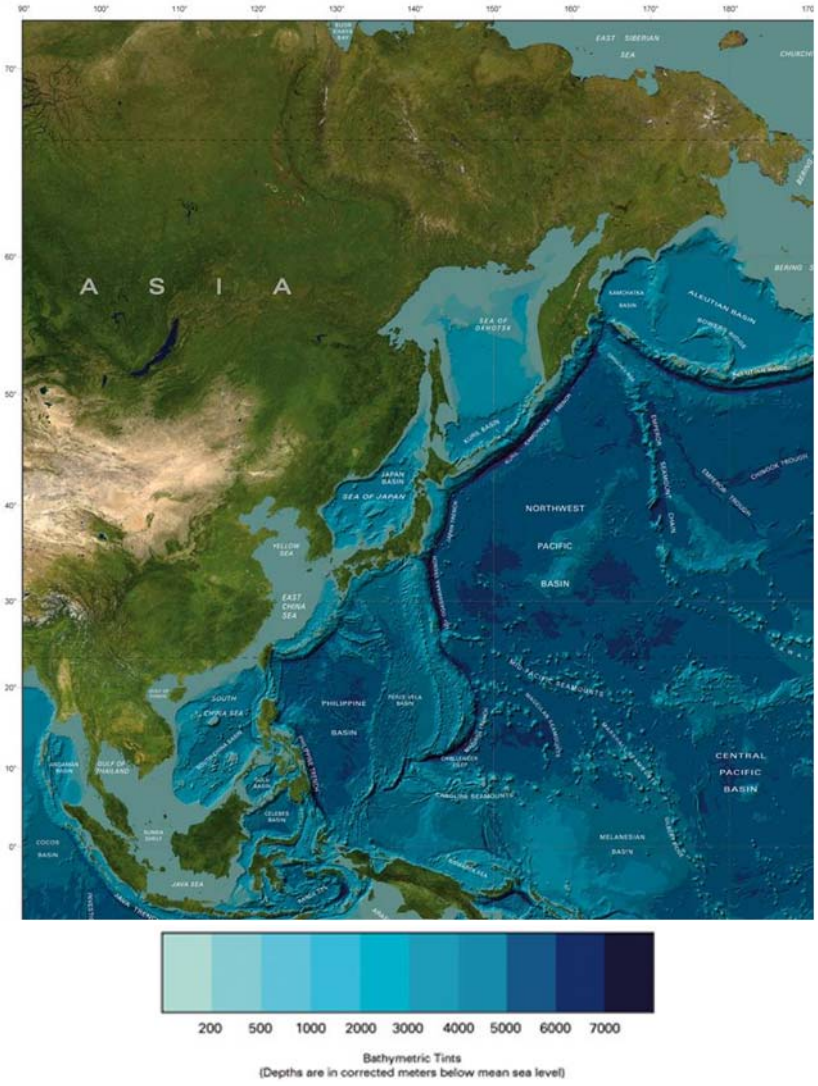


Figure 1. General bathymetric chart of the oceans and seas near Japan (image reproduced from the GEBCO world map)

The National Geophysical Data Center (NGDC) has published in 2010 a map of the total sediment thickness of the world's oceans and marginal seas as shown in Figure 2. The map was based on three main set of data: isopach maps (Divins, 2003), results from the Deep Sea Drilling Project and the Ocean Drilling Program and seismic reflection from NGDC. Figure 1 and Figure 2 clearly indicate that numerous areas within the Japan Sea and in the Northwest Pacific Ocean surrounding the Japanese archipelago have planar seafloor with underlying formations having sediment thickness ranging from several hundred feet to thousands of feet (sediment thickness in the Japan Basin and Northwest Pacific Basin around Japan are colored from light blue to green as shown in Figure 2).

Thus, from Figure 1 and Figure 2, several sites located within the Sea of Japan and the Northwest Pacific Ocean gather the conditions for technical assessments and studies of both the feasibility and viability of having carbon dioxide being captured onshore Japan from various industrial processes, then transported offshore to a deepwater or ultra-deepwater site and stored in sub-seabed sediments in order to mitigate part of the greenhouse gas emissions.

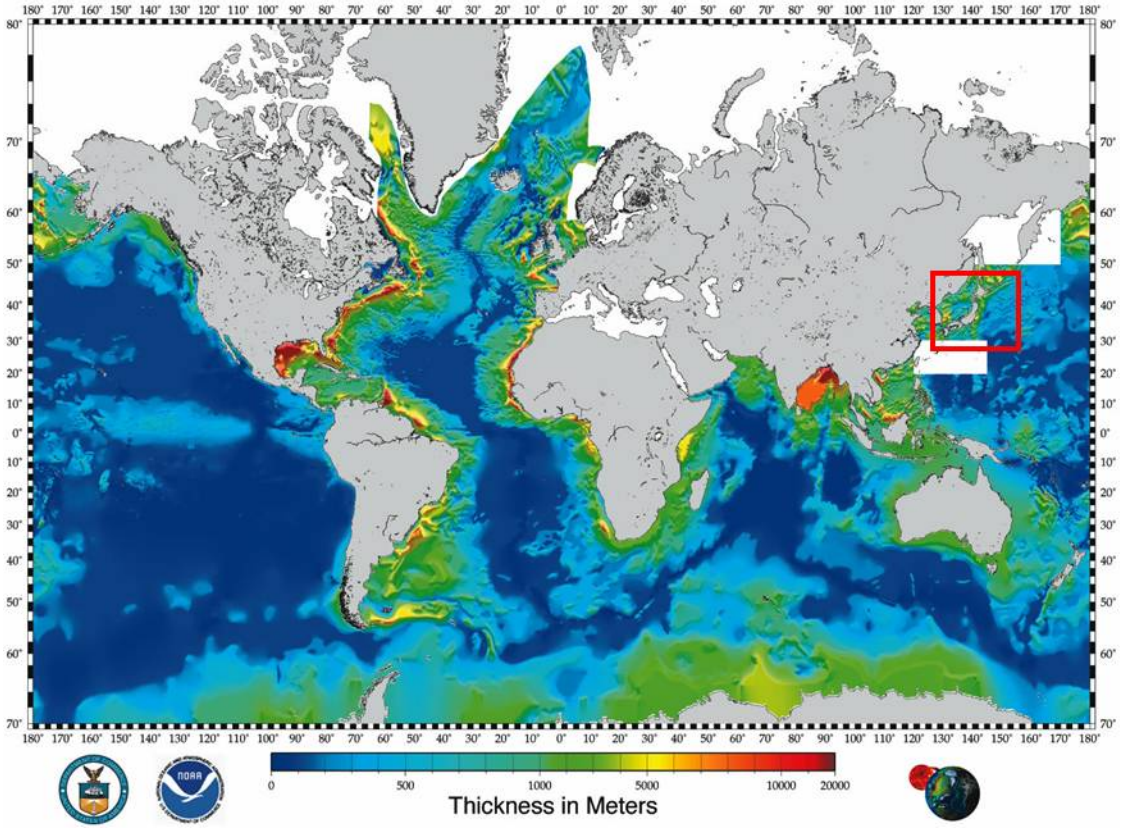


Figure 2. Total sediment thickness of the world's oceans and marginal seas (modified from National Geophysical Data Center)

CO_2 Capture, Transportation and Injection Offshore Japan

CO_2 is often captured and separated from several industrial processes but is vented to the atmosphere because of the absence of rules and the lack of tax incentives. Three main processes are currently used to isolate the CO_2 from the industrial gas stream: pre-combustion, post-combustion and oxy-fuel combustion. For this feasibility study review, both pre-combustion and post-combustion capturing technologies where the CO_2 is separated from the gas stream using a physical or chemical sorbent are selected as efficient and cost effective methodologies.

For all of these three case studies located offshore Japan, once the CO_2 is captured and separated from various industrial processes such as power generation plants, cement manufactures, oil refineries, iron and steel facilities within the major cities (Yokohama-Tokyo, Saitama, Sendai, Sapporo, Nagoya, Osaka, Kobe and Kyoto notably), it is then stored momentarily in large CO_2 tanks. Then, carbon dioxide is transported from the different CO_2 tanks using onshore CO_2 pipelines. CO_2 pipelines can carry CO_2 in its supercritical phase (CO_2 pressure above 1070 psi and temperature greater than 88°F). Due to pressure drops along the route, recompression stations are needed to maintain the CO_2 pressure above 1070 psi if the distance from the CO_2 tanks to the CO_2 terminal near shore is too large (≥ 100 miles). Then, once the supercritical CO_2 has reached the terminal, two means of transportation to carry the CO_2 from the terminal onshore to the injection vessel offshore deepwater can be used. First, subsea pipelines lying on the seabed and connecting the subsea injection wellheads to the shore pipeline terminals. However, deepwater pipelines have not been installed yet in the Sea of Japan and the Northwest Pacific Ocean and would require significant capital expenditure to build a deepwater CO_2 pipeline network. Besides, because of the very high tectonic activity of the region, this mean of transportation is not identified for the moment as economically viable for CO_2 injection in deepwater formations.

CO_2 can also be transported by ship which appears to be the best solution for CO_2 sequestration in the ultra-deepwater regions near the Japanese archipelago. Transporting CO_2 via ships requires much less capital expenditure than transporting it through ultra-deepwater pipelines and also provides greater flexibility, thus allowing the CO_2 to be conveyed to multiple injection sites. The ships use semi-pressurized and semi-refrigerated tankers to keep the CO_2 in its liquid phase similar to the ones used to transport liquefied petroleum gas (LPG) or liquefied natural gas (LNG) products and can carry CO_2 volumes over 500,000 ft^3 which can allow the injection of more than 100 million tons each year if the CO_2 injection site is at relatively close distance to shore such as the ones located in the deepwater parts of the Sea of Japan or offshore the island of Shikoku in the Northwest Pacific Ocean. In addition, even in the case of the third injection site located in the Northwest Pacific Ocean farther from the Japanese nearest shore, at least 10 million tons of CO_2 could be injected each year around each site which would still constitute the largest CO_2 sequestration project in the world. In addition, once the CO_2 transportation vessel has reached the injection site, the CO_2 can be offloaded using a fluid discharge process similar to the one used with oil tankers where the CO_2 ship is connected to an offshore buoy that is tethered to a submerged turret located in the center of the injection vessel and where the CO_2 is temporary stored before CO_2 injection operations can start.

Then, the selection of a drilling, storing and injection structures for operations in the deep waters off the coasts of Japan must take into account an exhaustive number of parameters to ensure the success of CO_2 sequestration in deepwater formations. The essential criteria to be considered are water depth, metocean conditions (waves, current, wind), operating period (year-round), offloading and storage of the CO_2, injection capability and rate. Currently, in the deepwater industry, semi-submersible ships (5th or the latest 6th generation) are most often the best suited option when operating in potential rough water and weather environments such as the Northwest Pacific Ocean and the Sea of Japan because of their better stability in pitch, roll and heave as compared to ship shaped vessels. In addition, other concepts such as circular shaped vessels that could perform both drilling and injection operations and that could also store the CO_2 after being offloaded from the CO_2 transportation ship are currently under development and could provide a more economic alternative (Srinivasan et al, 2008).

Therefore, for the injection operations in the near future, a large 5th or 6th generation semi-submersible vessel is chosen as the optimum choice to drill a shallow wellbore and inject CO_2 in ultra-deepwater sediments of either the Sea of Japan or the Northwest Pacific Ocean. Although being a major obstacle until recently, drilling and injecting CO_2 in water depths greater than 9,000 ft is within the limits of the deepwater oil and gas drilling industry (the 10,000 ft water depth frontier was passed in 2003 by one of the latest 5th generation drill-ship in the Gulf of Mexico and the new world record was recently established offshore India in over 10,200 ft of water in April 2011). Moreover, in order to conduct drilling and injection operations through over 10,000 of water column, riser type systems need to be employed.

Drilling and injection risers are both key components to ensure the structural and pressure integrity when the drilling fluids or the injected CO_2 are being carried from the vessel floor to the seafloor. Drilling risers can be described as vertical rigid pipes with added buoyancy modules on the outside and injection risers are flexible pipes, steel catenary risers (SCRs) or composite material pipes. The design of drilling and injection risers is very critical for these ultra-deepwater CO_2 injection operations because risers undergo repeated external and internal loading from the waves, ocean currents, drilling fluids and injected liquefied CO_2.

Thus, front-end engineering analyses have already been carried out and documented for steel drilling and injection riser systems for a CO_2 sequestration feasibility study in 11,000 feet of water in the Gulf of Mexico (Pilisi et al., 2010) but new results are provided in Table 1 for different drilling and injection riser configurations (steel risers, titanium risers, hybrid: steel+titanium risers, slim steel risers and composite injection risers). The results from these static and dynamic analyses are for a dynamically positioned semi-submersible vessel first located in about 9,000 feet of water off the coasts of Japan as shown in Table 1 and then for a vessel located in 18,000 feet of water in the Northwest Pacific Ocean as shown in Table 2.

The analyses were segmented for three different environmental loadings (calm, moderate and rough weather conditions) and show that new riser system materials, notably titanium alloy drilling riser and composite injection riser systems or slim riser systems made of steel (16 in. diameter instead of 21 in. in diameter), are required to ensure that drilling and injection operations can be conducted successfully year-round and even during ‘extreme’ weather conditions (100 year event). However, if the injection site is located in water depths less than 10,000 feet, in a calm weather area, ‘conventional’ steel drilling and injection riser systems can be deployed to conduct CO_2 sequestration operations.

Besides, if the water depths are greater than 10,000 feet like, for instance, the injection site N°3 located in the Northwest Pacific Ocean (see Figure 6) where the seafloor is located at approximately 18,000 feet below the mean water level, then, only new riser systems made with titanium alloy for calm, moderate and rough weather areas or slimmer steel tubes (16 in. diameter), only for calm weather locations, could be used to inject the CO_2 in ultra-deepwater sediments as shown in Table 2.

Table 1. Dynamic response of drilling and injection risers for injection sites N°1 and N°2

SITE N°1 (Japan Sea) and SITE N°2 (Offshore Shikoku - Northwest Pacific Ocean)						
DRILLING AND INJECTION RISERS: STATIC AND DYNAMIC RESPONSES						
RISER TYPE / MATERIAL TYPE	Drilling Riser (21 in. OD Steel Material)	Drilling Riser (21 in. OD Titanium Material)	Drilling Riser (21 in. OD Hybrid: Steel + Titanium Material)	Drilling Riser (16 in. OD Steel Material)	Injection Riser (8 in. OD Steel Material)	Injection Riser (8 in. OD Composite Material)
CONDUCT OPERATIONS WITH CALM WEATHER	√	√	√	√	√	√
CONDUCT OPERATIONS WITH MODERATE WEATHER	X	√	√	√	X	√
CONDUCT OPERATIONS WITH ROUGH WEATHER	X	√	X	√	X	√

Table 2. Dynamic response of drilling and injection risers for injection site N°3

SITE N°3 (Northwest Pacific Ocean N40° E150°)							
DRILLING AND INJECTION RISERS: STATIC AND DYNAMIC RESPONSES							
RISER TYPE / MATERIAL TYPE	Drilling Riser (21 in. OD Steel Material)	Drilling Riser (21 in. OD Titanium Material)	Drilling Riser (21 in. OD Hybrid: Steel + Titanium Material)	Drilling Riser (16 in. OD Steel Material)	Injection Riser (8 in. OD Steel Material)	Injection Riser (8 in. OD Titanium Material)	Injection Riser (8 in. OD Composite Material)
CONDUCT OPERATIONS WITH CALM WEATHER	X	√	X	√	X	√	X
CONDUCT OPERATIONS WITH MODERATE WEATHER	X	√	X	X	X	√	X
CONDUCT OPERATIONS WITH ROUGH WEATHER	X	√	X	X	X	√	X

Ultra-deepwater CO_2 Injection Well

The injector wells that need to be drilled in the deepwater sediments have to be very shallow wells if the liquid CO_2 needs to remain permanently stored through the presence of a 'gravitational' trap (i.e. high pressure and low temperature) within the sub-seabed formations. Thus, this makes the drilling operations and wellbore design very peculiar and somewhat much easier as compared to 'conventional' deepwater drilling which target is to produce hydrocarbons trapped relatively deep underground (several thousand feet below the seafloor).

Moreover, the injection of liquid CO_2 in the deepwater sediments which is a critical component in the overall success of the carbon dioxide sequestration process can be achieved by pumping this fluid down through an injection riser and into a cased wellbore. Thus, the bottom part of the injection casing needs to be perforated to enable the liquid CO_2 to enter the sub-seabed sediments. Also, for these applications, the perforation interval ranges between 10 feet and 30 feet depending on the injection depth, the permeability of sediments and the thickness of the sediments. In addition, the last casing string or injection casing can be set either within or just below the zone where the CO_2 density is greater than 'air+seawater' pore-fluid. Hence, the injection casing setting depth can range between 300 feet and 3,000 feet depending on the geothermal gradient in the sediments. In addition, it has been estimated that a 7 inch injection casing could convey CO_2 volumes averaging 3,000 tons per day (i.e. over 1 million tons per year). Besides, once the carbon dioxide leaves the wellbore at relatively shallow depths, the injection rate that may vary between 1,000 tons per day and 5,000 tons per day creates instantly a pressure increase in the vicinity of the wellbore, thus, allowing the liquid CO_2 to enter the pore-fluid constituted of both the void space and formation fluids and therefore starting the sequestration process.

Furthermore, the following conceptual wellbore configurations are proposed for the three locations off the coasts of Japan. Thus, two main configurations can be designed: one for shallow depth CO_2 injection and one for CO_2 injection relatively deep in the sediments. Figure 3 illustrates the two wellbore configurations: one for a shallow injection well (500 feet below the seafloor) and one for a deep injection well (3,000 feet below the seabed).

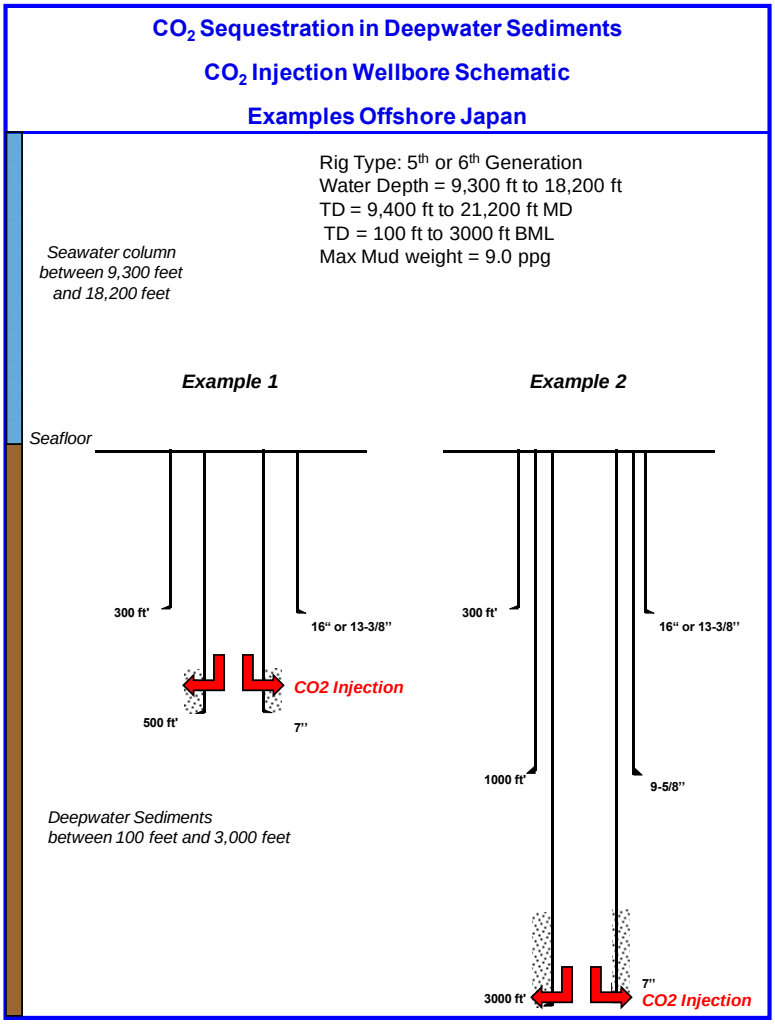


Figure 3. Examples for shallow and deep CO_2 wellbore configurations

In the first configuration, when the CO_2 is injected right below the seabed at about 300 feet in the sediments to benefit from the lowest temperature conditions reigning at that depth, a 13.375 inch conductor/surface casing string can be set at about 100 feet below the seafloor and then a 7 inch injection casing can be set at about 300 feet in the deepwater sediments.

Also, a second configuration considers the case of a slightly deeper injection well where a 16 inch or a 13.375 inch conductor/surface casing can be set at about 300 feet below the seafloor depending on the use of a 16 inch steel slim drilling riser systems to drill the shallow sediments; and a 7 inch injection casing which can be set at 500 feet in the sediments as shown on the left part of Figure 3.

In the third wellbore configuration, if the CO_2 is injected at greater depths; for instance at 3,000 feet below the seabed, then a 16 inch conductor/surface casing can be set at about 300 feet; then, a 9.625 inch intermediate casing can be set at about 1,000 feet; and then the 7 inch injection casing can be installed at about 3,000 feet within the deepwater sediments as shown on the right part of Figure 3. However, at this depth, the sediment temperature has a direct effect on the CO_2 behavior changing its phase from liquid to supercritical and therefore becoming much lighter than the surrounding pore-fluid. In addition, similar to the design of CO_2 injection wellbores used to enhance recovery and production of oil and natural gas wells, corrosion resistant materials that include chrome in their metallurgical properties need to be used for the injection casing and down-hole components.

CO_2 Sequestration in Deepwater Sediments Overall Process

Figure 4 illustrates the entire CO_2 sequestration in deepwater sediments process where the carbon dioxide can be captured and separated from any industrial process and then, once stored temporarily in CO_2 tanks, is transported in its supercritical phase to the nearest shore where it is unloaded to a refrigerated CO_2 ship where the CO_2 remains in its liquid phase (stage 1 on Figure 4). Then, the CO_2 is unloaded offshore to a deepwater floating unit which can be a drilling, storing and/or injection facility or only composed of a drilling unit for the drilling phase to drill the injection well and; afterward, a storing and injection unit can move on location to perform the injection operations (stage 2 on Figure 4). At that time, the liquid CO_2 is ready to being injected into the deepwater sub-seabed geological formations using injection riser pipes and can, thus, migrate within the deepwater sediments through perforations set in the 7 in. injection casing string (stage 3 on Figure 4).

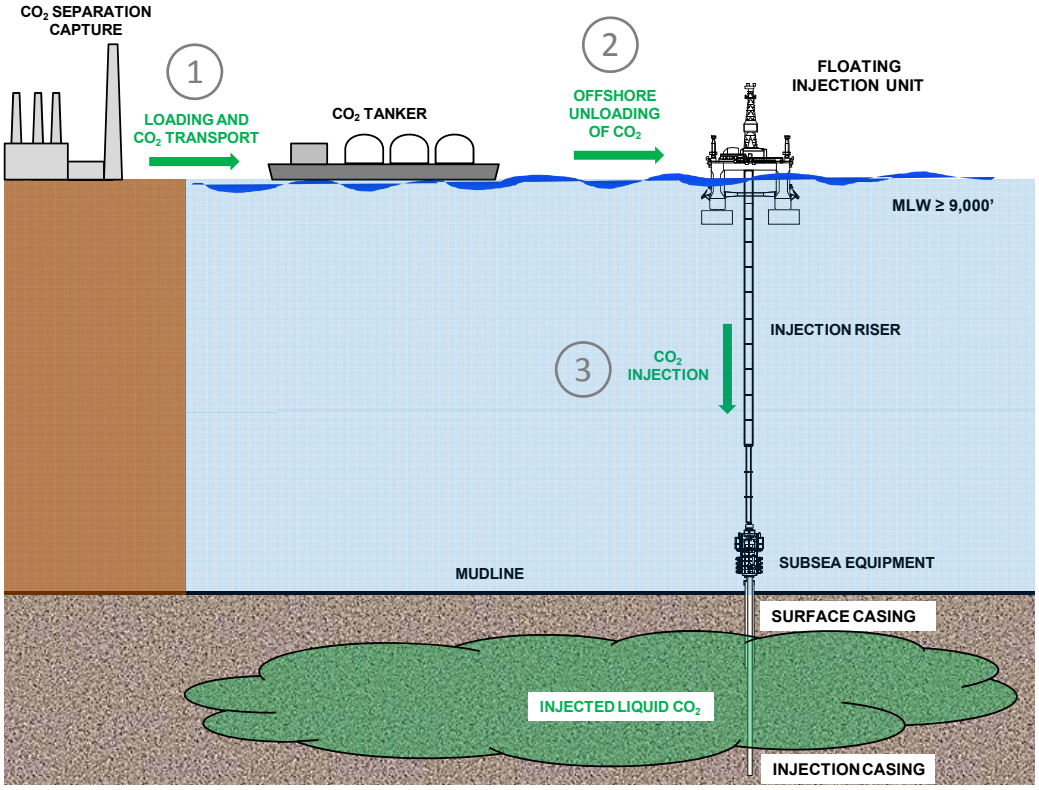


Figure 4. CO_2 sequestration process from capture to injection in deepwater offshore sediments

CO_2 Injection Site Characteristics

Nevertheless, drilling in an ultra-deepwater environment and injecting CO_2 in the first few hundred feet of sediments, to benefit from a 'gravitational' trapping mechanism that ensure that the liquid CO_2 will remain sequestered indefinitely requires that the injection sites be selected cautiously considering a number of critical parameters that may greatly influence the reservoir engineering viability studies. Thus, the potential well-site locations chosen for this paper should have the following characteristics and known data:

a. Water depth greater than 9,000 feet.
b. Sediment thickness greater than 100 feet.
c. Geothermal gradient (30°C/km to 120°C/km depending on local heat flow in the deepwater regions).
d. Sediment permeability (ranging from a few µD to 100 mD).
e. Sediment porosity (ranging from 40% to 90%).
f. Nearest port located not too far from the deepwater site (≤ 200 miles).

Since seawater, pore-water fluids and CO_2 properties change as a function of depth (i.e. pressure) and temperature, seawater and geothermal gradients data are critical parameters to be computed in the reservoir models. Thus, based on data from the

National Oceanic and Atmospheric Administration (NOAA) related to world's oceans and deep seas temperature and a breadth of peer reviewed articles and books (Pickard et al., 1990), the mean seawater temperature ranges from 40°F to 80°F at surface, depending on the latitude and seasons, to an asymptotic value of 35°F-39°F once it reaches about 4,000 feet water depth, and then, the seawater temperature remains constant down to 9,000-18,000 feet water depth. In addition, as depth increases, temperature of the sediments is a function of the thermal properties of the rock and regional tectonics and, hence, changes CO_2, pore-fluid and seawater densities.

Extensive literature is available concerning scientific expeditions that have been conducted with the Deep Sea Drilling Project (DSDP between 1968 and 1983), the Ocean Drilling Program (ODP between 1985 and 2004) and the Integrated Ocean Drilling Program (IODP since 2004) in order to study the ocean and deep sea basins of the world and notably numerous boreholes have been drilled, monitored and cored in different deepwater regions of the Sea of Japan and the Northwest Pacific Ocean around Japan. From the different cores recovered, permeability and porosity data have notably been measured and have been used for the reservoir simulations detailed in this paper.

Also, detailed studies have been recently been published specifically focusing on geothermal gradient and heat flow data in the Japan Sea and Northwest Pacific area (Figure 5: Tanaka et al, 2004). Figure 5 shows a wide range in heat flow values of the lithosphere for the Sea of Japan and the Northwest Pacific Ocean over short horizontal distances. Since the heat flow (Q) is the product of the thermal conductivity of the rocks or sediments (c) by the geothermal gradient (dT/dz) of the rocks or sediments as shown in Equation 1, and assuming that the thermal conductivity of the sediments that can range between 1.5 and 6 $W.m^{-1}.K^{-1}$ depending on the sediment type, is constant, the orders of magnitude of the geothermal gradients in the Japan Sea and Northwest Pacific Ocean are similar to the ones of the heat flow (up to a factor of 10 between dark blue dots having values of 20 $mW.m^{-2}$ and red dots having values of 200 $mW.m^{-2}$).

Equation 1: $$Q\,(W \cdot m^{-2}) = c\,(W \cdot m^{-1} \cdot K^{-1}) \times \frac{dT}{dz}\,(K \cdot m^{-1})$$

This unique lithospheric thermal regime can be explained because the Japanese archipelago is a volcanic arc bordered by a trench and a subduction region. Therefore, geothermal gradients in the Northwest Pacific Ocean are lower on the north-east side of the main islands of Honshu and Hokkaido with values ranging between 1°F/100 feet and 3°F/100 feet and geothermal gradients become greater on the South part of the Northwest Pacific Ocean with values between 4°F/100 feet and 8°F/100 feet. On the other hand, in the Sea of Japan, geothermal gradients range, in general, between moderate values of 2°F/100 feet in the Yamato Basin and close to the Russian Federation shores and higher values of 6°F/100 feet in the Japan Basin.

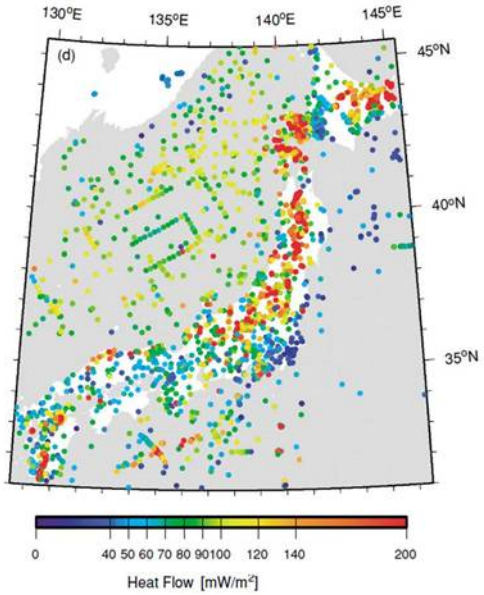


Figure 5. Heat flow in and around the Japanese archipelago (taken from Tanaka et al., 2004)

Besides, scientific drilling results from leg 31 (DSDP, Watanabe et al., 1972), legs 127 and 128 (ODP, Langseth and Tamaki, 1992, Pisciotto et al., 1992) located in the Sea of Japan and leg 131 (ODP, Hyndman et al., 1991) located in the Northwest Pacific Ocean show relatively high geothermal gradients with values of 6-7°F/100 feet.

Reservoir Simulations for 3 Injection Sites Offshore Japan

To model the liquid CO_2 plume mobility once it is injected in the sediments, a three dimensional equation of state compositional reservoir simulator (GEM^{TM}) has been used with the additional greenhouse gas module (GHG). The numerical simulations have been run after building a three dimensional reservoir model with a 25x25x60 Cartesian grid blocks for heterogeneous deepwater sediments with the injection well positioned in the center at grid (13, 13) and completed in the lower grid blocks. Through three case studies taking place in the Sea of Japan (site N°1) and the Northwest Pacific Ocean (sites N°2 and N°3) as shown in Figure 6, the CO_2 is injected in the deepwater sub-seabed formations, at water depths ranging between 9,300 feet and 18,200 feet, in deepwater sediments located at about 328 feet below the seafloor with a rate of 5,000 tons per day for 25 years. Then, the injection operations are stopped but the simulations are still run for another 225 years (i.e. 250 years since the beginning of the injection operations). Finally, the results of the numerical models for three different injection sites show the variations in CO_2 concentration in the sediments pore-volume and the immobility or migration of the CO_2 as a function of sediment temperature, pressure, permeability and porosity.

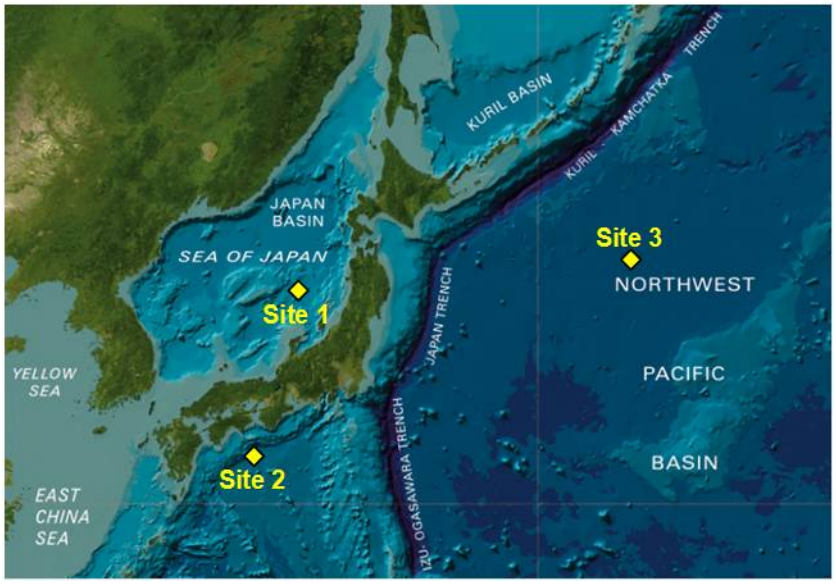


Figure 6. Location of the CO_2 injection sites offshore Japan (modified image reproduced from the GEBCO world map)

The main parameters that have been used from the CO_2 injection sites located around the coasts of Japan are summarized in Table 3. The three sites are situated around the Japanese archipelago: one being located in the Japan Sea, injecting CO_2 in lithogenous sediments in about 9,300 feet of water with a high geothermal gradient averaging 6°F/ 100 feet; the second site is positioned in the Northwest Pacific Ocean, offshore the main island of Shikoku in pelagic sediments with similar water depth and geothermal gradient than the Japan Sea injection site; and the third site is placed farther out in the Northwest Pacific Ocean (N40°, E150°) in water depths greater than 18,000 feet and with a relatively low geothermal gradient averaging 2°F/ 100 feet within the first few hundred feet of the pelagic sediments.

Table 3. Summary of the main parameters used in the numerical simulations for the three CO_2 injection case studies

LOCATION	Sediment Types	Porosity	Permeability	Water depth (Feet)	Sediment Thickness (Feet)	Pressure at Seabed (psi)	Temperature at Seafloor	Average Geothermal Gradient	Injection Rate (Tons/day)
Japan Sea	Lithogenous (Basaltic/Doleritic)	60% to 80%	0.1 mD to 40 mD	9281	328	4150	35°F	6°F/100 ft	5000
Offshore Shikoku, Pacific Ocean	Pelagic Clay (Calcareous Ooze)	42% to 66%	7.5 mD to 18.5 mD	9850	328	4405	35°F	6°F/100 ft	5000
Pacific Ocean (N40°, E150°)	Pelagic Clay (Siliceous Ooze)	67% to 83%	2 µD to 1.6 mD	18158	328	8120	35°F	2°F/100 ft	5000

The data used as input in the numerical simulations are petrophysical property measurements of the deepwater sediments that come from samples that have been collected by the Deep Sea Drilling Project (Pisciotto et al., 1992; Hill et al., 1993) and the Ocean Drilling Program (Kanazawa et al., 2001) in boreholes located nearby each of the three injection sites. Based on some of these data, multiple porosity and multiple permeability models have been generated using unconditional Gaussian geostatistical simulations for the numerical simulations.

Simulation Results

The numerical simulations for site N°1 considers a reservoir constituted of deepwater sub-seabed lithogenous sediments with basaltic deposits having a total thickness of 328 feet (100 meters) located in the Japan Sea in 9,281 feet of water. The temperature and pressure at the seafloor are respectively 35°F and 4,150 psi and the average geothermal gradient in the first few hundred feet of sediments is as high as 6°F per 100 feet. For site N°2, the reservoir is composed of deepwater pelagic sediments with a total thickness of 328 feet (100 meters) located in the Northwest Pacific Ocean, near the island of Shikoku, Japan in 9,850 feet of water. The temperature and pressure at the seafloor are respectively 35°F and 4,405 psi and the average geothermal gradient in the first few hundred feet of sediments is also 6°F per 100 feet. The reservoir simulations for site N°3 also located in the Northwest Pacific Ocean, near the coast of Japan compute a similar reservoir model containing pelagic sediments with a total thickness of 328 feet (100 meters). However, since site N°3 is located in 18,158 feet of water, the at the seafloor is greater (8,120 psi) and the geothermal gradient in the first few hundred feet of sediments for this area is much lower than the previous injection sites with 2°F per 100 feet.

For site N°1, the results of the numerical simulations show the global mole fraction of the injected CO_2 in the deepwater lithogenous sediments after a timeframe of 250 years as shown in the upper part of Figure 7. It can be seen that the CO_2 migrates both in the lateral and vertical directions but remains gravitationally trapped within the sediments on the lower part of the numerical model after some time. Indeed, the relatively high geothermal gradient of 6°F per 100 feet has a direct effect on the migration of the CO_2 plume. Thus, it can be seen that CO_2 mole fraction takes values of 1.0 (red color on Figure 7) in the vicinity of the injection zone representing about 2x2x6 grid blocks. Also, it appears that CO_2 concentration values in the upper and lateral parts of the sediments model are greater than zero which indicates that the CO_2 has migrated upward and then stayed gravitationally trapped in the bulk volume and can, therefore, be considered as immobile (light blue and green colors on Figure 7). In addition, CO_2 mole fraction values in the farthest lateral and the most upper parts of the numerical model are null which shows that the CO_2 has not reached these locations in the sediments even after 250 years, staying trapped within the pore-volume and, hence, proves the viability of the sequestration method (blue color on Figure 7).

Additionally, the results of the numerical simulations for site N°2 also show the global mole fraction of the injected CO_2 in the deepwater pelagic sediments after a period of time of 250 years. Similarly to the previous reservoir for site N°1, the CO_2 migrates both in the lateral and vertical directions because of the relatively high geothermal gradient of 6°F per 100 feet present within the deepwater sediments; but, as well, remains trapped within the pelagic formations after reaching a gravitational equilibrium on the bottom part of the reservoir model as illustrated in the middle of Figure 7. Again, CO_2 mole fraction takes values of 1.0 (red color on Figure 7) in the vicinity of the injection zone for a slightly larger volume than for site N°1, representing about 3x3x7 grid blocks. Likewise, CO_2 concentration values in the upper and lateral parts of the numerical model are greater than zero which indicates that the CO_2 plume has moved in the vertical and lateral directions and then has been gravitationally trapped in the bulk volume (light blue and green colors on Figure 7). Also, CO_2 mole fraction values in the farthest lateral and the most upper parts of the numerical model are null which shows that the CO_2 has not migrated to these locations, remaining immobile within the pore-volume, thus demonstrating the benefit of this carbon dioxide sequestration solution (blue color on Figure 7).

For the third CO_2 injection site, the results of the numerical simulations show the global mole fraction of the injected CO_2 in the ultra-deepwater lithogenous sediments of the Northwest Pacific Ocean after an elapsed time of 250 years (bottom part of Figure 7). As compared to the two previous numerical simulations, it can be seen that the CO_2 has not migrated as much in both the vertical and horizontal directions remaining trapped for the most part within the sediments. This is can be explained because of the presence of a relatively low geothermal gradient of 2°F per 100 feet in the rocks, at this location. Thus, CO_2 mole fraction is very high values of 1.0 (red color on Figure 7) in the vicinity of the injection zone for a larger volume than for the previous case studies (sites N°1 and N°2) representing about 4x4x8 grid blocks. Furthermore, CO_2 mole fraction values in the upper and lateral parts of the reservoir model are not null indicating again that the CO_2 is trapped within the

deepwater sediments. In addition, for this case, it can be discussed that the lower permeability of these sediments (ranging between 2μD and 1.6mD) has little effect on the CO_2 sequestration and that the gravitational trap is still functional. This is probably due to the very high formation pressure (i.e. 18,200 feet water depth), relatively high porosity of the deepwater sediments and the high injection rate of 5,000 tons of CO_2 per day which all combined may counterbalance the effects of low permeability deepwater sediments on the viability of the CO_2 injection concept.

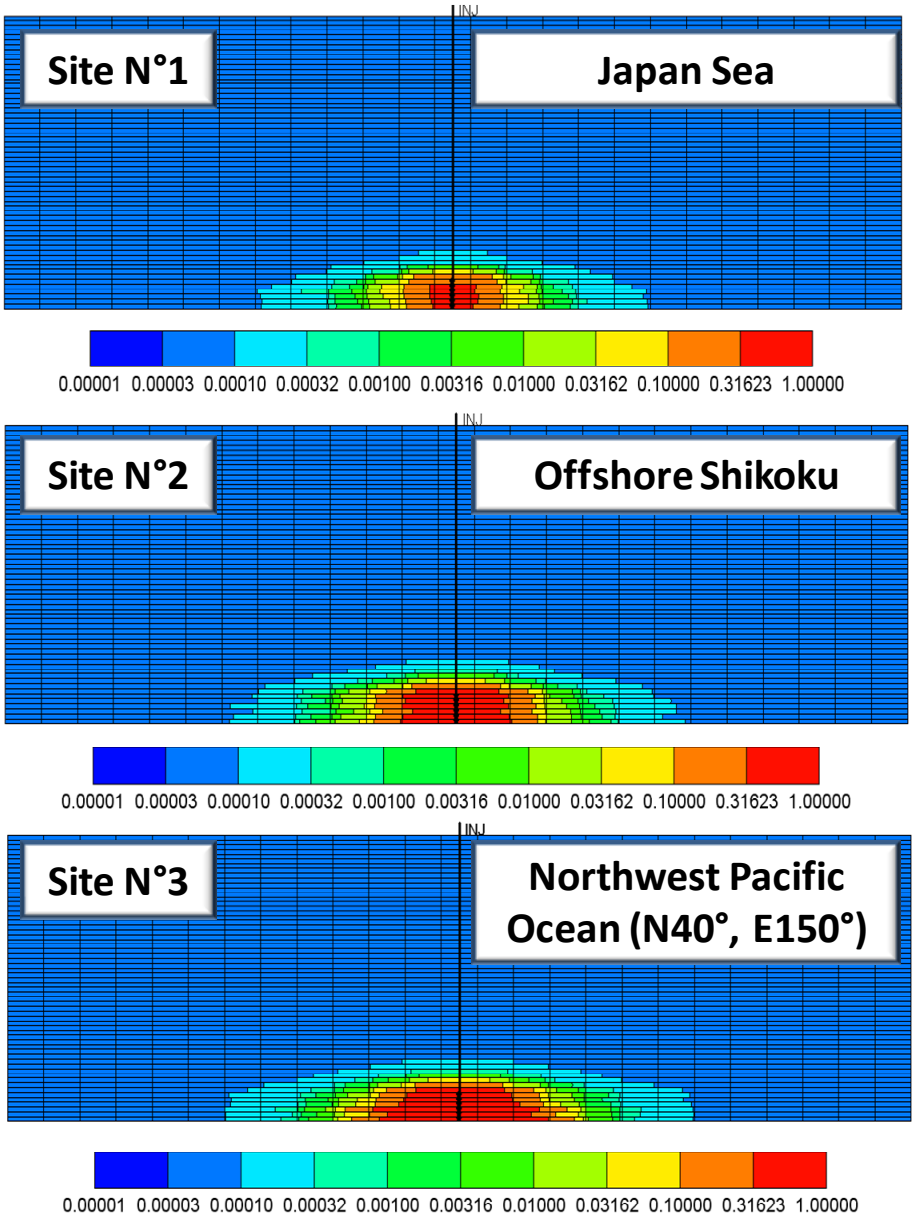


Figure 7. CO_2 mole fraction profile in deepwater sediments located offshore Japan

Moreover, an additional numerical simulation has been run for the case where CO_2 is injected at 3,000 feet below the seabed, in the deepwater sediments of the Japan Sea (i.e. water depth equals 9,281 feet). However, at these conditions of temperature and pressure (P=5,450 psi and T=200°F at 3,000 feet) CO_2 changes its state from liquid phase to supercritical phase. Therefore, supercritical CO_2 migrates relatively rapidly upwards until changing back its phase from supercritical to liquid; and finally reaches a depth below the seafloor where the temperature in the sediments is much cooler (≤80°F) than the temperature at the injection depth (i.e. 200°F) as shown in Figure 8. Hence, at this relatively shallow depth, liquid CO_2 becomes gravitationally trapped within the sediments due to its liquid phase density being heavier than the surrounding pore-fluid density.

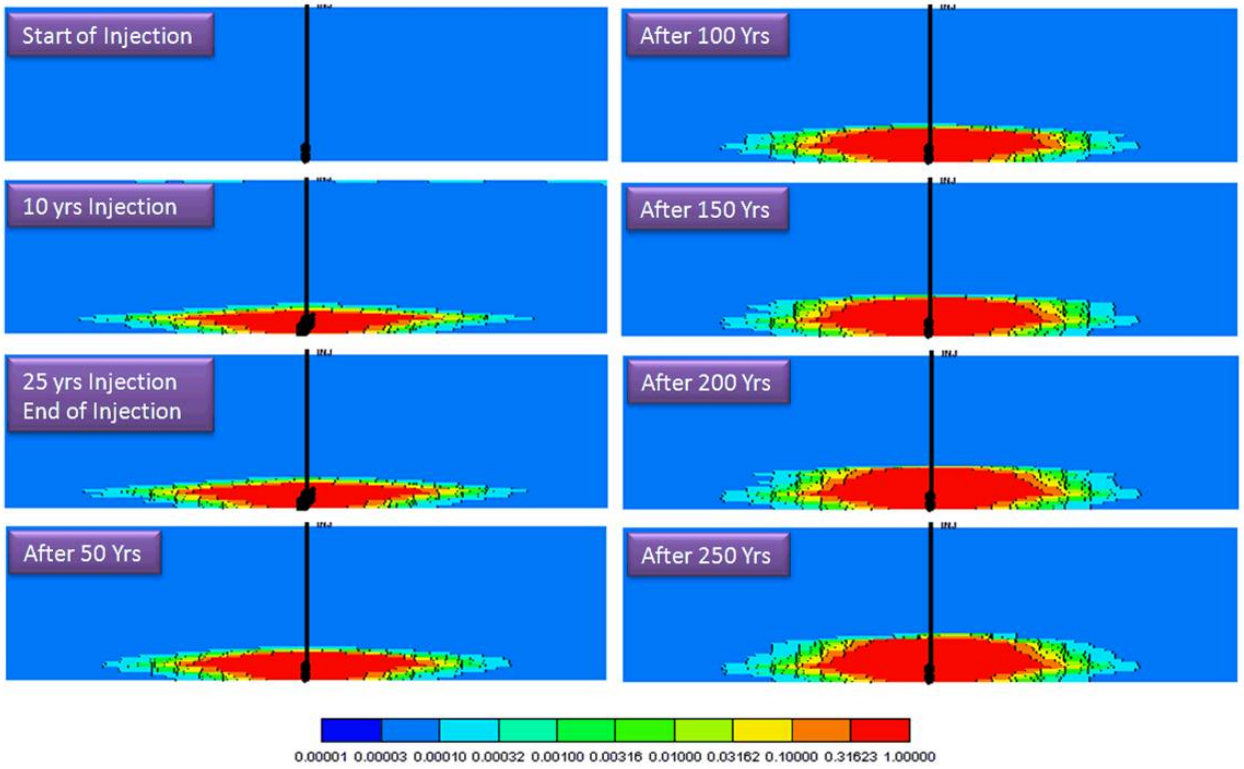


Figure 8. Evolution of the CO_2 plume between 3,000 feet below the seabed and the seafloor in the Japan Sea

Conclusions

Through three feasibility studies located offshore Japan (in the Japan Sea, offshore Shikoku in the Northwest Pacific Ocean and farther northeast in the Pacific Ocean), the main processes, technical requirements and technologies that are currently available, or may be needed, to capture, transport and inject the CO_2 successfully in deepwater sub-seabed formations have been reviewed. Thus, both capturing technologies and CO_2 transportation by ship are presently at a mature stage. In addition, drilling shallow CO_2 injection wellbores in water depths ranging between 9,300 feet and 18,200 feet are within the capability of the newest 5th and 6th generation drilling vessels if the main components (i.e. drilling and injection riser systems) are made of titanium alloys, composite materials or lighter steels. In addition, at the moment storing and/or injection vessels are not yet available to conduct CO_2 injection operations in ultra-deepwater environments.

Additionally, for each of the three CO_2 injection sites, results from computer models show that the storage of liquid CO_2 as an immobile fluid gravitationally trapped within the sediments pore-volume is viable because of the conditions of pressures (high) and temperatures (low) that are present in these deepwater sub-seabed formations when the CO_2 is injected relatively shallow under the seabed. Indeed, the injection depth seems to play a critical role in the mobility of the liquid CO_2 once it is injected into the deepwater sediments as illustrated when carbon dioxide is injected in the deepwater sediments at 3,000 feet below the seafloor. Thus, at relatively high sediment temperatures (≥80°F), CO_2 changes its phase to become supercritical and therefore has lighter density than the sediments pore-fluid. Hence, the local geothermal gradient value influences greatly the migration of the CO_2 plume in the sediments. Moreover, overall, under certain deepwater sediment parameters (i.e. low temperature, high pressure, relatively high porosity and permeability), it has been shown in this paper that CO_2 sequestration in deepwater sub-seabed formations is a viable and much safer technique for storing CO_2 than the structural cap rock located on top of an aquifer within which supercritical CO_2 is injected into or even when anthropogenic CO_2 is injected in depleted oil and gas fields for storage.

In addition, even though theoretically this CO_2 sequestration solution should not be sensitive to any structural geology or regional tectonics since the denser CO_2 stays at the bottom of the 'air+seawater' pore volume, more numerical simulations accounting for geomechanical perturbations will be required to demonstrate the integrity of the gravitational trap in earthquake sensitive areas such as the Japanese archipelago.

Furthermore, additional reservoir simulations are required to demonstrate that the injection of liquid CO_2 will not change the ocean chemistry because of the displaced formation fluid present originally within the deepwater sediments. Also, CO_2 hydrates formation may act either as an additional trapping mechanism or a limiting factor for CO_2 injection and could therefore extend or reduce the number of potential candidates for CO_2 sequestration in deepwater sediments.

Finally, CO_2 storage in deepwater sub-seabed formations could be used on a global scale to mitigate greenhouse gas emissions to permanently store the captured CO_2 from various industrial processes; and thus constitute a viable alternative especially for densely populated areas, seismic sensitive regions such as Japan.

Acknowledgements

The authors would like to thank Blade Energy Partners for permission to publish this paper and the Society of Petroleum Engineers to invite us to present during the Carbon Management Technology Conference.

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